Unstitched: level design workbook

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Every number here is measured from the 13 shipped toys (Assets/_Project/Scripts/Core/Toys.cs) by Tune.Report, which plays the game's own rules in Unity. Nothing is estimated. Blue cells are inputs (the sample size and the confidence); the formulas update when you change them.

Sheet	What it shows
Levels	Every toy: its shape (parts, rounds, colours, spools, sewing, layers, tray), how many positions it has and how many are lost, the solver's fewest rounds in the tray, how two bots fared with a margin of error, and how its spool order compares with other orders.
Order check	Each toy against the one before it: is it clearly harder, too close to call, or out of order, given the margin of error?
Curve	Summaries as formulas over the Levels sheet: with and without layers, and how the measures move together (correlations).
Inputs	The games per measure and the z for the margin. Change them and every margin and verdict follows.

Measures

Measure	Meaning
Positions	Every different state the toy can reach from the start (which rounds are off, the spools, the tray).
Lost share	The share of positions from which no win is left: found by walking back from every win.
Fewest in tray	The solver's line that sends the fewest rounds to the tray. 0 means a perfect order needs no waiting.
Random player	Pulls any legal round. 1,000 games a toy (seed 7). Says how forgiving a toy is.
Careful player	Pulls a colour with an open spool whenever one shows, otherwise at random; sees only what is showing. Says how fair a toy is without peeking.
Random wins, 100	The same player over 100 games, as the tuner measured when the orders were chosen.
Orders	Every different order of the toy's spools. With 400 or fewer, all are measured; otherwise a sample of 80 winnable orders found by swapping two spools at a time from the chosen one (seed 11). Same tray as the game.

The bots are not people. Unstitched has not been playtested by first-time players yet; these numbers set the order until it is.

Inputs: the sample and the margin

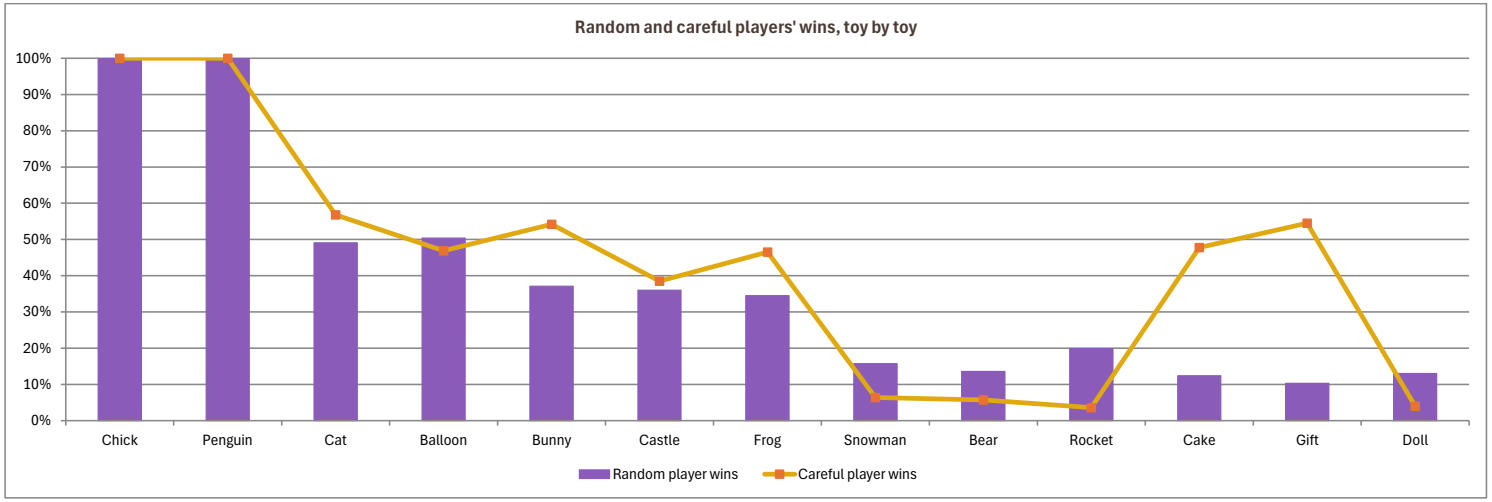
Blue cells are inputs; the Levels and Order check sheets read them.

Input	Value	Note
Games per measure	**1,000**	Tune.Report plays 1,000 random games a toy.
z for a 95% margin	**1.96**	Margin = z × √(p(1 − p) / games).

Every toy, measured

From Tune.Report over the shipped toys. Green = more forgiving, red = harder. Grey columns are formulas.

Level	Toy	Parts	Rounds	Colours	Spools	Sewn joins	Layers	Free parts at start	Tray places	Positions	Lost share	First pulls	First pulls that lose	Fewest in tray (solver)	Wins taking whole strands	Random player wins	Careful player wins	Careful minus random	Random wins, 100 games (as tuned)	95% margin	Spool orders	Winnable orders measured	Of those, can't be lost	More forgiving than the chosen	Share more forgiving
1	Chick	3	12	2	3	1	0	2	2	75	0%	2	0	0	yes	100%	100%	0%	100%	±0.0%	3	3	100%	0	0%
2	Penguin	3	16	3	4	1	0	3	0	58	0%	2	0	0	no	100%	100%	0%	100%	±0.0%	12	9	89%	0	0%
3	Cat	6	24	3	6	4	0	5	2	395	14%	5	0	0	yes	49%	57%	+8%	53%	±3.1%	30	25	88%	22	88%
4	Balloon	4	28	5	7	3	0	2	1	63	29%	2	0	0	yes	50%	47%	-4%	45%	±3.1%	1,260	80	80%	74	93%
5	Bunny	6	32	4	8	4	1	4	3	410	16%	4	0	0	yes	37%	54%	+17%	40%	±3.0%	840	80	55%	69	86%
6	Castle	8	28	5	7	7	0	4	2	331	38%	4	0	0	yes	36%	39%	+3%	33%	±3.0%	840	80	68%	63	79%
7	Frog	7	32	5	8	6	1	3	1	585	5%	3	0	0	yes	35%	47%	+12%	31%	±2.9%	3,360	80	80%	79	99%
8	Snowman	6	32	6	8	4	1	3	2	146	47%	3	0	0	yes	16%	6%	-9%	22%	±2.3%	6,720	80	96%	80	100%
9	Bear	8	32	6	8	7	1	4	2	288	28%	4	0	0	yes	14%	6%	-8%	23%	±2.1%	6,720	80	84%	78	98%
10	Rocket	6	24	4	6	5	0	5	2	880	40%	5	0	2	yes	20%	4%	-16%	18%	±2.5%	180	84	48%	74	88%
11	Cake	8	36	7	9	7	2	4	3	217	22%	4	0	2	yes	12%	48%	+35%	15%	±2.0%	90,720	80	61%	79	99%
12	Gift	7	40	6	10	7	1	4	3	321	10%	4	0	2	yes	10%	55%	+44%	12%	±1.9%	226,800	80	31%	75	94%
13	Doll	7	40	5	10	8	2	3	2	129	28%	3	0	8	yes	13%	4%	-9%	11%	±2.1%	75,600	80	65%	78	98%


Random and careful players' wins, toy by toy
100%
90%
80%
70%
60%
50%
40%
30%
20%
10%
0%
Chick
Penguin
Cat
Balloon
Bunny
Castle
Frog
Snowman
Bear
Rocket
Cake
Gift
Doll
Random player wins
Careful player wins

Order check: is each toy harder than the last?

A change counts only when it is bigger than the two margins together (formulas over the Levels sheet).

Level	Toy	Random wins	The toy before	Its random wins	Change	Combined margin	Verdict
2	Penguin	100%	Chick	100%	0%	±0.0%	the same
3	Cat	49%	Penguin	100%	-50.9%	±3.1%	clearly harder
4	Balloon	50%	Cat	49%	+1.3%	±4.4%	too close to call
5	Bunny	37%	Balloon	50%	-13.3%	±4.3%	clearly harder
6	Castle	36%	Bunny	37%	-1.1%	±4.2%	too close to call
7	Frog	35%	Castle	36%	-1.5%	±4.2%	too close to call
8	Snowman	16%	Frog	35%	-18.8%	±3.7%	clearly harder
9	Bear	14%	Snowman	16%	-2.1%	±3.1%	too close to call
10	Rocket	20%	Bear	14%	+6.3%	±3.3%	**clearly easier: out of order**
11	Cake	12%	Rocket	20%	-7.5%	±3.2%	clearly harder
12	Gift	10%	Cake	12%	-2.1%	±2.8%	too close to call
13	Doll	13%	Gift	10%	+2.7%	±2.8%	too close to call

How to read it: 100 games an order were enough for the tuner to rank one toy's orders, but two neighbouring toys can sit within each other's margin. Measured with 1,000 games, the rocket (level 10) is clearly easier than the bear before it: either swap them or pick a harder rocket order, then measure again.

Curve: layers, and what moves together

Formulas over the Levels sheet.

Toys	How many	Average random wins	Average careful wins	Average careful minus random
Without a layer	6	59%	58%	-2%
With a layer (from level 5)	7	20%	31%	+12%
Levels 5 to 13 without a layer	2	28%	21%	-7%

What moves together (Pearson correlation over the 13 toys)

Pair	r
Rounds and the random player's wins	-0.89
Positions and the random player's wins	-0.38
Lost share and the random player's wins	-0.63
Free parts at start and the random player's wins	-0.47
Layers and careful minus random	0.36

How to read it: a negative r means the more of the first, the fewer games a careless player wins. With 13 toys a correlation is a hint, not proof. The careful player's gap shows where hidden colours punish pulling whatever fits: a negative gap means thinking ahead (or peeking) matters.